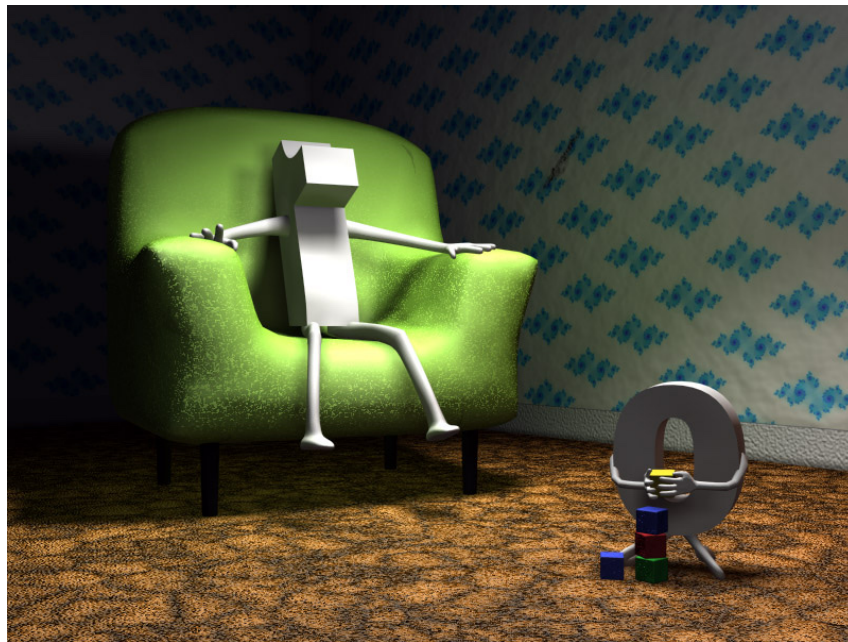


Chapter 1. INTEGERS

1. 2 Divisibility and Prime Factorisation

1.2.4 Prime Factorisation



Today we will prove a result that might sound familiar to you already. It is known as the

PRIME FACTORISATION THEOREM

PRIME FACTORIZATION THEOREM.

- (1) Every integer $n \geq 2$ is a product of (one or more) primes.
- (2) The factorization is unique up to the order of the factors.
That is, if $n = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$, where all the p_i and q_j are primes, then $r = s$ and the q_j can be relabelled so that $p_i = q_i$ for all $i = 1, 2, \dots, r$.

Proof of (2). Suppose that (2) does not hold. That means that n admits two different factorisations into primes. By the Well-Ordering Axiom we can choose $m \geq 2$ to be the smallest integer admitting two different factorisations into primes:

$$m = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$$

Notice that m is not a prime and so $r, s \geq 2$.

Euclid's Lemma

$$p_1 \mid m = q_1 q_2 \dots q_s \Rightarrow p_1 \mid q_j \text{ for some } j.$$

Reordering the q_i 's (if necessary) we can assume that $p_1 \mid q_1$. From here we get that $\underline{p_1 = q_1}$.

$$\frac{m}{p_1} = p_2 p_3 \dots p_r = q_2 \dots q_s \quad \begin{array}{l} \text{integer} < m \\ \text{with two different} \\ \text{factorisations into} \\ \text{primes.} \end{array}$$

That contradicts the fact that n was the smallest, so (2) must hold. $\square$

THEOREM 1.22. Let n be an integer with prime factorization

$$n = p_1^{n_1} p_2^{n_2} \dots p_r^{n_r},$$

where the p_i are all distinct primes and $n_i \geq 1$ for each i . Then the positive divisors of n are precisely the integers d of the following form:

$$d = p_1^{d_1} p_2^{d_2} \dots p_r^{d_r}, \text{ with } 0 \leq d_i \leq n_i \text{ for all } i.$$

Definition 1.23. Let $n_1, n_2, \dots, n_r$ be positive integers. The **least common multiple** $\text{lcm}(n_1, n_2, \dots, n_r)$ of $n_1, \dots, n_r$ is the (positive) common multiple of $n_1, \dots, n_r$ that is a divisor of any other common multiple of $n_1, \dots, n_r$.

$$\text{lcm}(a, b) \cdot \text{gcd}(a, b) = ab.$$

Theorem 1.25 and
Corollary 1.26

If a and b are positive integers, and

$$a = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_r^{b_r} \quad \text{then}$$

$$\text{lcm}(a, b) = p_1^{m_1} p_2^{m_2} \dots p_r^{m_r},$$

$$\text{gcd}(a, b) = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r},$$

where $m_i = \max\{a_i, b_i\}$ and $k_i = \min\{a_i, b_i\}$ for all i

Example 1.24. Find $d = \gcd(12, 20, 18)$ and $m = \text{lcm}(12, 20, 18)$.

$$12 = 2^2 \cdot 3 = 2^2 \cdot 3^1 \cdot 5^0$$

$$20 = 2^2 \cdot 5 = 2^2 \cdot 3^0 \cdot 5^1$$

$$18 = 2 \cdot 3^2 = 2^1 \cdot 3^2 \cdot 5^0$$

$$\underline{d = 2^1 \cdot 3^0 \cdot 5^0 = 2}$$

$$\underline{m = 2^2 \cdot 3^2 \cdot 5^1 = 180}$$